

Common protocols used by home automation systems

X10. One of the oldest protocols in this space, it has evolved from power line to wireless. Cannot claim to be very robust in terms of speed or communications, and hence not too popular.

ZigBee. A happening technology, IEEE 802.15.4 (aka Zigbee) is a low-power communication technology that makes mesh networks to help devices communicate on par with each other.

Z-Wave. Another wireless home protocol, owned by Sigma Designs, which makes and sells chips for Z-Wave-capable products.

Insteon. Combines wired power line and wireless communications to form mesh networks for the devices to communicate when in proximity. If one mesh fails, the others can take over. Insteon was developed by SmartLabs, and is compatible with X10.

Wi-Fi. Wi-Fi needs no introduction. Although it is power and bandwidth hungry, experts feel that the use of Wi-Fi in home automation can definitely not be ignored due to its popularity and ubiquity of access points.

households, there is a growing need for intelligent surveillance of residential properties. Home owners are looking for solutions that can help them clearly identify intruders, monitor their homes while they are on the go," says Sudhindra Holla, country manager, Axis Communications, India, which offers several advanced security products and solutions.

Indeed, security appears to be the biggest motivation for implementing smart home solutions in India. Builders and real-estate developers are installing holistic surveillance systems that help to monitor large residential campuses.

Traditional measures such as locks and extra doors have been replaced by high-end burglar alarm systems and security products with integrated remote monitoring software. Security guards are also supported by intelligent CCTV surveillance systems.

There is a growing demand for cameras with smarter analytics, especially vandal-proof cameras and video motion detectors. "The latest tech trends unfolding on home security are razor-sharp detail on video, motion detection and tamper-proof cameras. In India, we see a higher adoption of thermal cameras, cameras with high definition (HD) motion sensors and facial recognition software," says Holla. A thermal network camera uses thermal imaging, allowing users to detect people, objects and incidents in complete darkness and difficult conditions such as smoke, haze, dust and light fog.

Likewise, smartphones and video-capable networking have also given

home security a boost. Real-time access to recorded visuals is a reality today, and there are several affordable smartphone apps that make good use of this. "Home automation technology has become an integral part of nearly every new alarm panel that hits the market these days. This will enable mobile users to monitor their home from any corner of the world 24x7," says Holla.

Surveillance-as-a-service (SaaS) is another growing trend in India. This helps home users switch from low-end analogue cameras to network-based IP surveillance easily and at a low cost.

Globally, there are several wonderful examples of security products that can be deployed quite easily by the user itself. WeMo offers a range of motion sensors and other products that you can install for various home automation purposes including security.

Canary recently launched a smart home security device that is armed with an HD video camera and multiple sensors that track everything from motion to room temperature, sound and activity to help keep your home safe.

Another example would be Piper, which is a home security and automation device that combines panoramic video, Z-Wave home automation and environmental sensors into a single elegant product that you interact with on your smartphone or tablet.

Electricity bills give a pleasant surprise

"While currently the demand for smart home products in India is limited to safety and surveillance/monitoring,

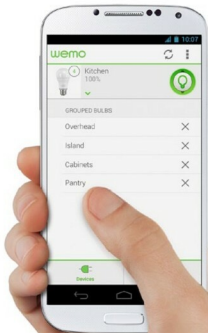
we believe this will change in the future because technology will allow people to use resources more efficiently, improving convenience, saving money and facilitating a greener lifestyle," says Guru Ganesan, president and MD, ARM India. "We are already seeing automation used fairly widely for monitoring energy and water usage to help reduce a household's environmental footprint and electricity bills."

There are several smart technologies today that would help a home reduce its energy footprint, and some high-end builders are including such technologies to conserve power used by lighting, elevators and other common amenities. So are home owners.

There are many ways in which home automation can conserve power. For example, lights and HVAC systems can be switched off automatically the moment your car leaves the garage. Lights can switch on and off as people enter or leave a room.

Two-way meters and smart thermostats can help people understand and alter their energy usage.

Cloud-based services can help utilities to analyse, understand and recommend ways to reduce power consumption at user locations.



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